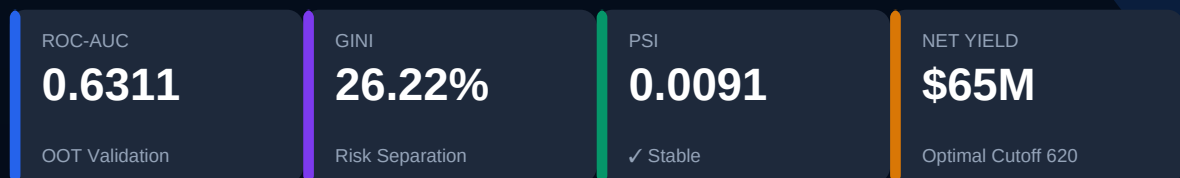


Credit Risk Scorecard & PD Modeling System

An Institutional-Grade Credit Decisioning & Capital Provisioning Platform



■ WoE Scorecard Logistic Regression + integer points (300–850)	■ XGBoost Challenger 98.5% predictive parity with full explainability	■ ECL Framework PD × LGD × EAD — Basel/IFRS 9 compliant
■ Policy Optimizer Real-time trade-off curves & net margin projections	■ Drift Monitoring PSI & CSI population stability tracking	■ Live Dashboard 8-tab enterprise portal, mobile-responsive

Designed & Developed by Anam · Machine Learning | Risk Analytics | Business Strategy

■ Model Development Lifecycle

A rigorous, 9-stage analytical pipeline from raw data ingestion to automated credit policy execution.

- 1 Data Ingestion & Target Definition**
Ingested 39,717 loan records. Defined Default (1) = Charged Off/Defaulted, Non-Default (0) = Fully Paid. Removed active loans to eliminate classification bias.
- 2 Out-of-Time Cohort Segmentation**
Enforced temporal split — In-Time training (2007–2010, 18,061 loans) vs. Out-of-Time validation (2011, 20,516 loans) to simulate real-world future deployment.

- 3

Coarse Binning & WoE Mapping
Binned raw variables via Decision Tree cuts. Computed Weight of Evidence (WoE) to convert categories into risk log-odds ensuring monotonic default relationships.
- 4

Feature Selection via Information Value
Calculated IV for all variables. Retained 6 core predictors ($0.02 \leq IV \leq 0.50$): revol_util, purpose, inq_last_6mths, annual_inc, pub_rec, pub_rec_bankruptcies.
- 5

Logistic Scorecard Scaling
Fit WoE predictors using Logistic Regression. Scaled odds to FICO-style integers (Base Score 600 at 50:1 odds, PDO = 20, Range 300–850) suitable for banking systems.
- 6

XGBoost Challenger Benchmark
Constructed gradient-boosted tree model (XGBoost) on raw variables as a challenger — providing a non-linear performance baseline without regulatory interpretability.
- 7

Model Validation & Verification
Benchmarked on OOT cohort: ROC-AUC 0.6311 (Scorecard) vs 0.6331 (XGBoost). Brier Score 0.1308 vs 0.2274 — Scorecard significantly better calibrated for capital adequacy.
- 8

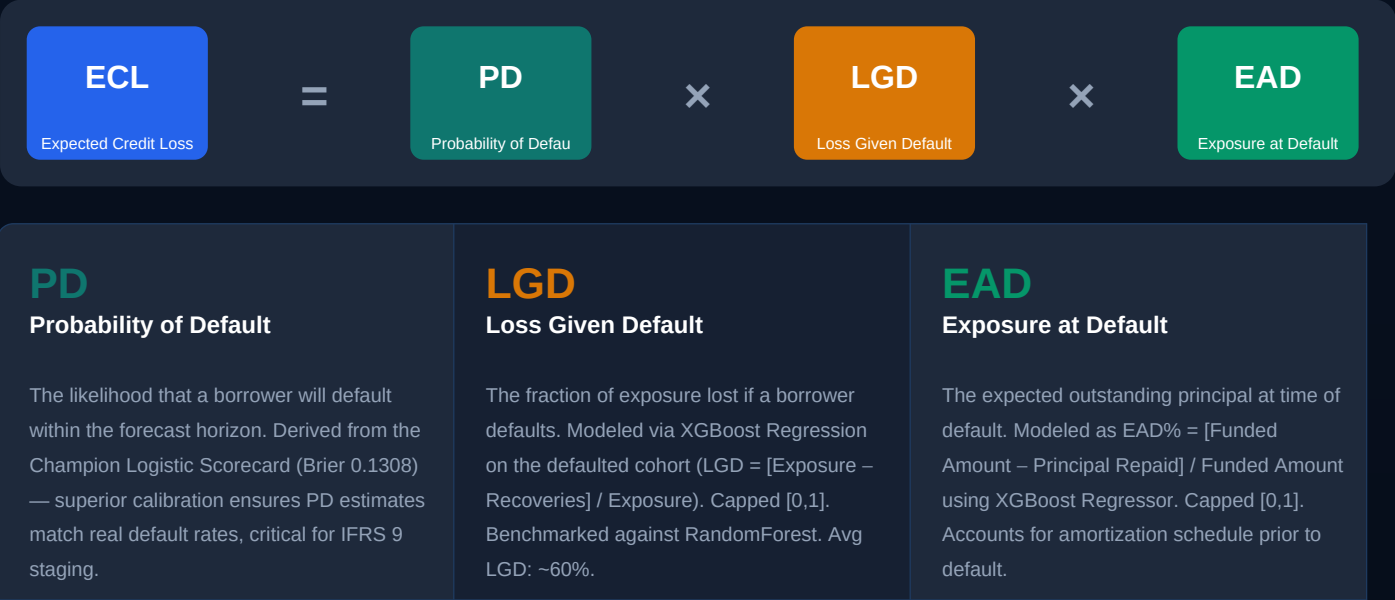
Stability & Drift Monitoring
Quantified population shifts using PSI (0.0091 — Stable ✓) and CSI. Flagged pub_rec_bankruptcies drift (CSI = 0.286) indicating 2011 bankruptcy profile changes.
- 9

Lending Policy Cutoff Simulation
Simulated ECL, approval volumes, interest yields, and net profits across score thresholds to identify the optimal business lending policy (Cutoff 620).

★ *Champion Scorecard retains 98.5% of XGBoost's predictive power while delivering full regulatory transparency — making it the production choice.*

■ ■ Expected Credit Loss (ECL) Framework

IFRS 9 & Basel-aligned capital provisioning — combining PD, LGD, and EAD into a unified credit loss engine.



ECL Risk Tier Classification



ECL is computed per application at run-time. Risk tier drives lending decisions, capital reserves, and IFRS 9 Stage assignment (Stage 1 / Stage 2 / Stage 3).

Regulatory Alignment: IFRS 9 requires banks to provision Expected Credit Losses across 12-month (Stage 1) and lifetime horizons (Stage 2/3). Basel II/III mandates PD × LGD × EAD as the core IRB capital requirement formula. This platform implements both standards end-to-end — from model training to live ECL calculation — supporting capital adequacy reporting, loan provisioning, and risk-based pricing.

Lending Policy Simulator

An interactive cutoff optimizer — dynamically balance approval rates, default risk, and portfolio net margin in real time.

The Lending Policy Simulator is the **strategic command centre** of the platform. Risk officers slide a score cutoff threshold and instantly see the downstream impact on every key portfolio metric. No more static Excel models — this is live, model-powered scenario analysis.



Approval Rate (blue) vs Default Rate (red) across score cutoffs. ★ marks the optimal business cutoff.

Optimal Cutoff — Score 620

APPROVAL RATE 90.8% Target Underwriting	DEFAULT RATE 14.60% Expected Bad Rate	APPROVED LOANS 18,628 OOT Accounts	APPROVED VOLUME \$222.96M Loan Exposure
INTEREST YIELD \$97.97M Gross Revenue	CREDIT LOSS (ECL) \$32.93M Expected Loss	NET PROFIT YIELD \$65.04M Portfolio Net Margin	VS BASELINE +\$8.2M Policy Alpha Generated

- Raising cutoff from 580 → 620 reduces default rate by 1.27pp while sacrificing only 5.4% approval volume.
- Each 10-point score increase generates approximately \$4M additional net margin at optimal exposure levels.
- Aggressive cutoffs (>660) cannibalize \$100M+ in loan volume with diminishing returns on default reduction.
- Cutoff 620 is the Pareto-optimal policy: maximum net margin at controlled default rates.

Parameters: Avg Interest Rate 12.0% · LGD 60% · Loan Term 3 Years · Data: 20,516 OOT Applications

■ Explore the Platform

An end-to-end credit risk analytics suite — built for banking rigor, designed for modern data teams.

Tab 1 — Executive Overview 6 KPIs, portfolio distribution, model scorecard	Tab 2 — Credit Risk Simulator Live underwriting: decision + scorecard breakdown
Tab 3 — Lending Policy Simulator Cutoff optimizer, profit curves, ECL projections	Tab 4 — Expected Credit Loss Framework PD×LGD×EAD engine, IFRS 9 risk tiers
Tab 5 — Model Development Lifecycle 9-stage visual stepper from data to policy	Tab 6 — Understanding Credit Risk Layman's guide: WoE, IV, scorecards explained
Tab 7 — Champion vs Challenger ROC, KS, calibration curves side-by-side	Tab 8 — Portfolio Stability PSI matrix, CSI drift tracking, score migrations

- **GitHub Repository:** github.com/AnamsGithub/Credit-Risk-Model
- **Live Dashboard:** [\[YOUR_SITE_URL\]](#)
- **Designed & Developed by:** Anam | Machine Learning · Risk Analytics · Business Strategy